EMPIRICAL BAYES

Bridging Bayesian and Frequentist Perspectives

TRANSITIONING TO EMPIRICAL BAYES

RECAP: BAYES' RULE

$$P(\theta|y) = \frac{P(y|\theta) \cdot P(\theta)}{P(y)}$$

> Updates prior beliefs P(θ) with data P(y|θ).

> Results in a refined posterior P(θ|y).

> Fundamental to Bayesian inference.

THE EB CHALLENGE

> Where does the prior P(θ) come from?

> EB estimates the prior directly from data.

> "Borrowing strength" across related cases.

> A bridge between Bayesian and Frequentist logic.

WHY USE EMPIRICAL BAYES?

OBJECTIVE PRIORS

Replaces subjective "expert opinion" with objective, data-driven evidence from the population.

INFORMATION SHARING

"Borrows strength" across multiple related parameters to improve individual estimates.

STABILITY

Provides more robust estimates for individual cases, especially those with limited data.

SHRINKAGE

Pulls extreme, noisy observations toward a more reliable group average for better accuracy.

THE CORE FRAMEWORK: HIERARCHICAL MODELING

HYPERPARAMETERS η

↓

PARAMETERS θ_i

↓

DATA y_i

LEVEL 2: THE PRIOR

Parameters θ_i are drawn from a common population distribution $G(\theta|\eta)$.

LEVEL 1: THE DATA

Observations y_i depend on individual parameters θ_i via likelihood $P(y_i|\theta_i)$.

THE EB TWIST

Instead of a "prior on the prior", we estimate η directly from all observed data y.

ROBBINS' NON-PARAMETRIC METHOD

THE CORE IDEA (1955)

Estimate the posterior mean without explicitly specifying or estimating the prior distribution's shape.

$$E(\theta|y) = (y+1)\frac{P(y+1)}{P(y)}$$

> Applies to Poisson likelihoods.

> Uses marginal frequencies P(y).

> Purely data-driven estimation.

EXAMPLE: ACCIDENT RATES

> Insurance company tracks accidents per customer.

> Individual rates θ are unknown.

> To predict θ for a customer with `y` accidents:

> Look at the ratio of people with `y+1` vs `y` accidents in the whole database.

> Naturally accounts for group-level evidence.

THE JAMES-STEIN ESTIMATOR: STEIN'S PARADOX

THE ESTIMATOR

Estimating a vector of means $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_p)$ from observations $\boldsymbol{y} = (y_1, \ldots, y_p)$, where $y_i \sim N(\theta_i, \sigma^2)$ and $p \geq 3$.

$$\hat{\boldsymbol{\theta}}_{JS} = \left(1 - \frac{(p-2)\sigma^2}{\|\boldsymbol{y}\|^2}\right)\boldsymbol{y}$$

> $\|\boldsymbol{y}\|^2 = \sum_{i=1}^{p} y_i^2$

> Shrinks MLEs towards the origin.

STEIN'S PARADOX & EB CONNECTION

> For $p \geq 3$, $\hat{\boldsymbol{\theta}}_{JS}$ always has lower Mean Squared Error (MSE) than the MLE ($\boldsymbol{y}$).

> Counter-intuitive: Even unrelated parameters benefit from pooling information.

> Can be derived as a Bayes estimator under a hierarchical model with estimated hyperparameters.

> Provides strong theoretical justification for shrinkage in Empirical Bayes.

THE POWER OF SHRINKAGE

WHAT IS SHRINKAGE?

> Moving individual estimates toward the global mean to reduce total error.

> **Stein's Paradox**: For 3+ parameters, the combined estimator is always better than individual MLEs.

> Essential for handling noisy data in high-dimensional settings.

BIAS-VARIANCE TRADE-OFF

> **Bias**: Introduced by pulling away from the raw data.

> **Variance**: Significantly reduced by leveraging group information.

> **Net Result**: Lower Mean Squared Error (MSE).

VISUALIZING THE PULL

Extreme observations are "shrunk" toward the population mean.

> Raw data (MLE) is often too noisy.

> EB provides a "regularized" estimate.

> The more noise in the data, the more we shrink.

PARAMETRIC EB: GAUSSIAN-GAUSSIAN

THE MODEL

$$y_i|\theta_i \sim N(\theta_i, \sigma^2)$$

$$\theta_i \sim N(\mu, \tau^2)$$

>> Likelihood: Observations y_i are noisy versions of θ_i.

>> Prior: Parameters θ_i come from a common normal population.

>> Hyperparameters: μ and τ^2 are unknown.

ESTIMATION & POSTERIOR

$$E[\theta_i|y_i] = w \cdot y_i + (1 - w) \cdot \mu$$

$$w = \frac{\tau^2}{\tau^2 + \sigma^2}$$

>> μ and τ^2 are estimated from the sample mean and variance of all y_i.

>> The weight w represents the "signal-to-noise" ratio.

>> Result: Individual estimates are shrunk toward the global mean μ.

PARAMETRIC EB: POISSON-GAMMA MODEL

LIKELIHOOD (DATA)

$$y_i|\theta_i \sim \text{Poisson}(\theta_i)$$

PRIOR (POPULATION)

$$\theta_i \sim \text{Gamma}(\alpha, \beta)$$

POSTERIOR

$$\theta_i|y_i \sim \text{Gamma}(\alpha + y_i, \beta + 1)$$

>> **Ideal for Count Data**: Used for modeling rates like disease incidence, accident counts, or web clicks.

>> **Conjugacy**: The Gamma prior is conjugate to the Poisson likelihood, making the posterior calculation exact and fast.

>> **Hyperparameter Estimation**: α and β are estimated from the entire dataset using MLE or Method of Moments.

>> **Handling Small Samples**: Effectively stabilizes estimates for groups with very small populations or zero counts.

CASE STUDY: BASEBALL BATTING AVERAGES

THE PROBLEM

> A player has 1 hit in 1 at-bat.

> Raw Average (MLE) = 1.000.

> Is he the greatest player in history?

> Small sample size leads to extreme, unrealistic estimates.

THE EB APPROACH

> Use the distribution of **all** players' averages as the prior.

> Most players hit between .200 and .330.

THE EB SOLUTION: SHRINKAGE

1.000 -----→ **.265**

Raw Data (MLE) shrunk toward League Mean

> The 1.000 average is "shrunk" toward the league mean (e.g., .260).

> The result is a much more realistic prediction of future performance.

> As the player gets more at-bats, the estimate moves closer to his raw data.

METHODOLOGICAL COMPARISON

FEATURE	FREQUENTIST (MLE)	EMPIRICAL BAYES	FULL BAYES
Prior	None	Estimated from Data	Pre-specified
Focus	Individual y_i	Group + Individual	Subjective + Data
Uncertainty	Confidence Intervals	Point Estimate of Prior	Full Posterior
Complexity	Low	Medium	High (MCMC)

Empirical Bayes offers a pragmatic middle ground, combining data-driven priors with Bayesian logic.

ADVANTAGES AND LIMITATIONS

ADVANTAGES

[+] **Data-driven Priors**: Objective and evidence-based.

[+] **Improved Estimates**: Shrinkage reduces total error.

[+] **Efficiency**: Faster than full MCMC hierarchical models.

[+] **Scalability**: Handles "many small problems" simultaneously.

LIMITATIONS

[-] **Model Risk**: Biased if the prior form is wrong.

[-] **Under-uncertainty**: Ignores hyperparameter error.

[-] **Not "Pure"**: A pragmatic hybrid, not strictly Bayesian.

[-] **Sample Size**: Needs enough data to estimate the prior.

Empirical Bayes is a powerful, pragmatic bridge between philosophies, offering robust estimation in the face of modern data complexity.